

EXPERIENCE

Xterra AI

Founding Research Scientist

Currently in stealth, stay tuned for more info!

San Francisco, CA

Nov 2025 – Present

Tribe AI

Technical Consultant (AI/ML)

Providing AI/ML expertise to companies from small to enterprise. Selected projects include:

- Built first-party, externally facing tutorials, workshops, and documentation for both Llama and GPT models. Includes tutorials and notebooks for fine-tuning, distillation, and on-prem deployment.
- Led a 6 month research project to incorporate RL into a series-A robotics company's tech stack. Focused on simulator selection (including generative sim), benchmark RL tasks, and RL fine-tuning.
- Built computer vision models for a custom generative jewelry business in 3 months to unblock public launch, including a VLM-based distillation pipeline.
- Designed and built a proof-of-concept for an agentic VLM to merge a large PE-backed aerial imagery company's existing aerial data and ML models with capabilities from few-shot multimodal LLMs.

San Francisco, CA

Dec 2023 – Nov 2025

Taro AI

Founder & CTO

Taro's platform used foundation models tuned for aerial imagery to help cities/arborists manage trees. Led engineering and product. Built Taro's software and models from 0-1, including remote sensing models and voice-AI systems.

San Francisco, CA

Dec 2023 – Dec 2024

Waymo

Staff Software Engineer, Research

Researched and published a novel scene tokenization system that took advantage of Waymo's unique datasets. Improved scene tokenization over lidar-only by an average of 10% on most metrics.

Senior Software Engineer, Perception

Built the lidar and camera perception systems for Waymo Via, the trucking and freight project. Translated six mature machine learning systems to the truck platform, and evaluated and re-trained each to driverless quality. Built a visual language query system to search Waymo's huge corpus of collected data (2B+ objects), used by engineering, product, and leadership teams.

San Francisco, CA

Feb 2023 – Nov 2023

Jul 2020 – Feb 2023

Cruise

Engineering Manager, Machine Learning

Led a nine-person team of researchers and engineers focused on improving object detection at Cruise. Built, maintained, and iterated a primary production object detector. Conducted applied research in object detection, tracking, and simulation. Reduced object detection related safety events by 40% in 6 months.

Senior Machine Learning Engineer

Tech lead and project owner: emergency vehicles, audio understanding, lidar detection (bird's-eye view). Researched deep learning models for computer vision, video classification, audio, lidar, tracking, and sensor fusion. Developed systems end-to-end, including hardware design, algorithm and infrastructure development, model deployment, and validation and testing. Built automated retraining and active learning systems to improve models automatically.

San Francisco, CA

Jan 2020 – Jul 2020

Feb 2018 – Jan 2020

Fitbit

Research Scientist

Spearheaded efforts to introduce deep learning to Fitbit R&D. Developed and validated TensorFlow-based networks on a variety of data modalities, which included building associated training infrastructure. Built machine learning tooling for model deployment both to the cloud as well as low-power microcontrollers.

Researched machine learning and signal processing methods for a variety of data, including images,

San Francisco, CA

June 2016 – Feb 2018

accelerometer/gyroscope, PPG, and audio. Wrote firmware that ships on products, including on the Fitbit Ionic and Versa. Acted as a technical mentor and leader, presenting at internal talks and mentoring interns.

University of Michigan Computer Science Dept.

Graduate Student Research Assistant

Ann Arbor, MI

August 2014 – June 2016

Investigated methods for using sensor data and novel deep learning networks to improve outcomes in cardiothoracic surgery. Researched advanced methods of static compiler optimization by applying deep sequence models to LLVM intermediate language.

Internships

DSP Intern, iZotope Inc.

Summer 2013

Software Intern, Digital Design Corp. (began at age 16)

Summer 2008/09/10/11/12

EDUCATION

MS (PhD dropout) in Computer Science, **University of Michigan**, Ann Arbor

2014 – 2016

BS in Electrical Engineering, **Tufts University**, Medford

2010 – 2014

SKILLS

Vision-Language Models; Generative AI; Agent-based Systems; Machine Learning; Foundation Models; Computer Vision/Perception; SciPy, NumPy, Pandas; PyTorch, JAX, TensorFlow; Reinforcement Learning

PUBLICATIONS

Harada, N., Ji, J., Yang, Z., Mu, N., Tang, H., Chen, K., & Zhou, Y. (2024). MoST: Multi-modality scene tokenization for motion prediction. *CVPR 2024*.

Zekany, S., **Harada, N.**, Rings, D., Laurenzano, M., Tang, L., & Mars, J. (2016). CrystalBall: Statically analyzing runtime behavior via deep sequence learning. *MICRO 2016*.

Harada, N., Saeed, M., Baveja, S., & Syed, Z. (2015). Evaluating the utility of a multi-factorial computational model and simplified multi-factorial risk score to predict postoperative atrial fibrillation following cardiothoracic surgery. *AHA Scientific Sessions*.

PATENTS

Scene Tokenization for Motion Prediction. Filed Nov 13, 2024 (pending).

Object detection based on three-dimensional distance measurement sensor point cloud data. US 11107227. Issued Aug 1, 2021